

Visco-Elastic Cycle Models

v6, corrected equations and verification status

DAM, with corrections by TWB / Claude

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*About this version: v5 edits the v4 equations where the implementation (see `relax_fault_ve.md` and `rsf_ve_theory_and_status.md` in `src/relax/`) suggested corrections; the two substantive ones are in sections 3 and 4 and are marked **Correction** in the text: the interaction kernel of the hereditary integral is the relaxation (step-response) function acting on the slip history, with no Dirac part, and the single-exponential form is exact only in the antiplane limit, the general kernels having three material relaxation times. Section 1 is unchanged apart from the μ versus ν typo and two closing remarks; the internal-variable architecture of sections 2 and 3 survives as written. These are offered as suggestions for DAM's review. The numerical evidence lives in the repository: the prescribed-slip testbed `relax_fault_ve` (with `relax_fault` implementing section 1 for comparison), the unit test `ve_check` (elastic-sum, held-out-pole, and relaxed-limit identities at rounding level on Okada, plane-strain, and triangular kernels), and `ve_laplace_check` (full time histories against Gaver-Stehfest numerical inversion), driven by `test_relax/run_ve_tests` and `test_relax/run_fault_relax_test`; the cited documents describe them.*

1 Equations

We will use tensor deviatoric plus volumetric decomposition for the stress tensor

$$\boldsymbol{\sigma} = \boldsymbol{s} + p\boldsymbol{I},$$

and the strain tensor

$$\boldsymbol{\epsilon} = \boldsymbol{e} + \frac{1}{3}\epsilon_v\boldsymbol{I},$$

where

$$\epsilon_v(\mathbf{u}) = \nabla \cdot \mathbf{u}, \quad \mathbf{e}(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) - \frac{1}{3} \epsilon_v(\mathbf{u}) \mathbf{I}.$$

We use the following definitions for the deviatoric stress $\mathbf{s}$ and the pressure (volumetric stress) via

$$\mathbf{s} = 2G\mathbf{e}, \quad p = K\epsilon_v,$$

where G is the shear modulus and K is the bulk modulus.

Coefficient helpers:

$$K = \frac{E}{3(1-2\nu)}, \quad G = \mu = \frac{E}{2(1+\nu)}$$

and

$$\mu = G, \quad \lambda = K - \frac{2}{3}G$$

and the Poisson's ratio is

$$\nu = \frac{3K - 2G}{6K + 2G} = \frac{\lambda}{2(\lambda + \mu)}.$$

The conservation of momentum ($\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}$) becomes

$$2G\nabla \cdot \mathbf{e}(\mathbf{u}) + K\nabla \epsilon_v(\mathbf{u}) = \mathbf{0}.$$

In the visco-elastic model we assume K is constant, e.g. viscous effects are confined to only the shear stress components, thus we have

$$\frac{1}{2G} \dot{\mathbf{s}} + \frac{1}{2\eta} \mathbf{s} = \dot{\mathbf{e}} = \mathbf{F}.$$

Discretize the above constitutive equation with implicit Euler, leading to

$$\frac{\mathbf{s}^{k+1} - \mathbf{s}^k}{\Delta t} + \frac{G}{\eta} \mathbf{s}^{k+1} = 2G \frac{\mathbf{e}^{k+1} - \mathbf{e}^k}{\Delta t}.$$

Doing some algebra (steps included to ensure I didn't screw it up) we have

$$\mathbf{s}^{k+1} + \Delta t \frac{G}{\eta} \mathbf{s}^{k+1} = 2G(\mathbf{e}^{k+1} - \mathbf{e}^k) + \mathbf{s}^k$$

and

$$1 + \Delta t \frac{G}{\eta} = \frac{\eta + \Delta t G}{\eta}$$

and

$$\mathbf{s}^{k+1} + \Delta t \frac{G}{\eta} \mathbf{s}^{k+1} = 2G\mathbf{e}^{k+1} - 2G\mathbf{e}^k + \mathbf{s}^k.$$

Finally we have

$$\mathbf{s}^{k+1} = \left(\frac{\eta}{\eta + \Delta t G} \right) (2G\mathbf{e}^{k+1} - 2G\mathbf{e}^k + \mathbf{s}^k).$$

Now the conservation of momentum $\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}$ becomes

$$\nabla \cdot \boldsymbol{\sigma}(\mathbf{u}^{k+1}) = \nabla \cdot \mathbf{s}(\mathbf{u}^{k+1}) + K \nabla \epsilon_v(\mathbf{u}^{k+1}) \quad (1)$$

$$= \nabla \cdot \left[\left(\frac{\eta}{\eta + \Delta t G} \right) (2G\mathbf{e}^{k+1} - 2G\mathbf{e}^k + \mathbf{s}^k) \right] + K \nabla \epsilon_v(\mathbf{u}^{k+1}) \quad (2)$$

$$= \mathbf{0} \quad (3)$$

thus we have

$$\nabla \cdot \left[\left(\frac{2G\eta}{\eta + \Delta t G} \right) \mathbf{e}^{k+1} \right] + K \nabla \epsilon_v(\mathbf{u}^{k+1}) = \nabla \cdot \left[\left(\frac{\eta}{\eta + \Delta t G} \right) (2G\mathbf{e}^k - \mathbf{s}^k) \right],$$

which we may write as

$$\nabla \cdot [2G'(\Delta t)\mathbf{e}^{k+1}] + K \nabla \epsilon_v(\mathbf{u}^{k+1}) = \nabla \cdot \left[\left(\frac{\eta}{\eta + \Delta t G} \right) (2G\mathbf{e}^k - \mathbf{s}^k) \right],$$

which looks like standard elasticity formulation with a modified shear modulus $G'(\Delta t)$ and a non-zero right side, where

$$G'(\Delta t) = G\alpha(\Delta t), \quad \alpha(\Delta t) = \frac{\eta}{\eta + \Delta t G}.$$

Stress Update

•

$$\mathbf{s}_{ve}^{k+1} = \text{dev}(\mathcal{F}(G\alpha(\Delta t), K))$$

•

$$\mathbf{s}^{k+1} = \mathbf{s}_{ve}^{k+1} - \text{dev}(\mathcal{F}(G\alpha(\Delta t), K)) + \alpha(\Delta t)\mathbf{s}^k$$

where $\mathcal{F}(\cdot)$ is the Green's function.

Better explanation

Let S^k denote the slip at time t^k and let

$$\boldsymbol{\sigma}^k = \mathcal{F}(G, K, S^k)$$

define the Green's function which maps slip S^k to stress $\boldsymbol{\sigma}^k$, that is $\mathcal{F}$ provides an analytic solution to

$$\nabla \cdot \boldsymbol{\sigma}^k = \mathbf{0}, \quad \text{jump}(\mathbf{u}) = S^k \quad \text{on } \Gamma_{\text{fault}}$$

where $\boldsymbol{\sigma}^k = 2G\mathbf{e}^k + K\epsilon_v^k\mathbf{I}$. Hence to compute the stress update on the fault Γ_{fault}

$$\boldsymbol{\sigma}^{k+1} = 2G'(\Delta t)\mathbf{e}^{k+1} + K\epsilon_v^{k+1}\mathbf{I} - 2G'(\Delta t)\mathbf{e}^k + \alpha(\Delta t)\mathbf{s}^k$$

we do

$$\boldsymbol{\sigma}^{k+1} = \mathcal{F}(G'(\Delta t), K, S^{k+1}) - \text{dev}(\mathcal{F}(G'(\Delta t), K, S^k)) + \alpha(\Delta t)\mathbf{s}^k.$$

This update requires storing the slip S^k from the previous time step along with $\mathbf{s}^k$ (the deviatoric stress). One might be tempted to store the result of $\text{dev}(\mathcal{F}(G'(\Delta t), K, S^k))$, however this would be unwise as Δt will change from step k to $k+1$.

It would be more explicit to define a step k dependent time step, i.e. Δt^k . Then $t^{k+1} = t^k + \Delta t^{k+1}$ so we would have

$$\boldsymbol{\sigma}^{k+1} = \mathcal{F}(G'(\Delta t^{k+1}), K, S^{k+1}) - \text{dev}(\mathcal{F}(G'(\Delta t^{k+1}), K, S^k)) + \alpha(\Delta t^{k+1})\mathbf{s}^k.$$

Note that in the very short time step limit $\Delta t^{k+1} \rightarrow 0$ the elastic response is recovered since

$$\alpha(\Delta t^{k+1}) \rightarrow 1, \quad G' \rightarrow G$$

and thus

$$-\text{dev}(\mathcal{F}(G'(\Delta t^{k+1}), K, S^k)) + \alpha(\Delta t^{k+1})\mathbf{s}^k \rightarrow -\text{dev}(\mathcal{F}(G, K, S^k)) + \mathbf{s}^k = 0$$

leaving behind

$$\boldsymbol{\sigma}^{k+1} = \mathcal{F}(G, K, S^{k+1}).$$

Remarks (v5). In the implementation of this scheme (`relax_fault`) the deviatoric step ratio is exactly $\alpha(\Delta t)$ on all kernel families tested, so the scheme does what the derivation above says. Its continuum limit decays all

deviatoric stress as e^{-t/τ_m} , however, which differs at leading order from the exact solution of section 4, whose slowest decay time is $\frac{3(1-\nu)}{1+\nu}\tau_m$ (a factor 1.8 at $\nu = 1/4$); eq I is therefore a one-relaxation-time approximation rather than a discretization that converges to the multi-pole solution as $\Delta t \rightarrow 0$. The volumetric part of eq I also inherits a Δt dependence from the Green's-function shortcut.

2 Alternative VE models

Isotropic 1D visco-elastic material would have a stress $\sigma(t)$ given by

$$\sigma(t) = \int_0^t E(t - \hat{\tau}) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau},$$

where ϵ is the strain and where the relaxation modulus $E(t)$ may be expanded as a Prony series given by

$$E(t) = E_\infty + \sum_{i=1}^N E_i \exp(-t/\tau_i).$$

In the above $\tau_i = \eta_i/E_i$ is the relaxation time of the i -th Maxwell element. Then sub-ing in we have

$$\sigma(t) = E_\infty \epsilon(t) + \sum_{i=1}^N h_i(t),$$

where $h_i(t)$ are the internal state variables (history variables) for stress. Specifically we have

$$h_i(t) = \int_0^t E_i \exp(-(t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau}$$

To find the state at $t^{k+1} = t^k + \Delta t$ we evaluate the history integral $h_i(t^{k+1})$ by splitting the time domain:

$$h_i(t^{k+1}) = \int_0^{t^k} E_i \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau} + \int_{t^k}^{t^{k+1}} E_i \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau}$$

which can be re-written as

$$h_i(t^{k+1}) = \exp(-\Delta t/\tau_i) \int_0^{t^k} E_i \exp(-(t^k - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau} + \int_{t^k}^{t^{k+1}} E_i \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau} \quad (4)$$

$$= \exp(-\Delta t/\tau_i) h_i(t^k) + \int_{t^k}^{t^{k+1}} E_i \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) \frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} d\hat{\tau} \quad (5)$$

so in general we may write the update as

$$h_i(t^{k+1}) = \exp(-\Delta t/\tau_i) h_i(t^k) + \Delta h_i.$$

Making the assumption that the strain rate is constant over each time step Δt , i.e.

$$\frac{\partial \epsilon(\hat{\tau})}{\partial \hat{\tau}} \approx \frac{\epsilon^{k+1} - \epsilon^k}{\Delta t} = \frac{\Delta \epsilon}{\Delta t}$$

we have

$$\Delta h_i \approx E_i \frac{\Delta \epsilon}{\Delta t} \int_{t^k}^{t^{k+1}} \exp(-(t^k + \Delta t - \hat{\tau})/\tau_i) d\hat{\tau} = E_i \tau_i (1 - \exp(-\Delta t/\tau_i)) \frac{\Delta \epsilon}{\Delta t}$$

leading to

$$h_i(t^{k+1}) = \exp(-\Delta t/\tau_i) h_i(t^k) + E_i \frac{\tau_i}{\Delta t} (1 - \exp(-\Delta t/\tau_i)) \Delta \epsilon$$

Then the discrete stress update is

$$\sigma^{k+1} = E_\infty \epsilon^{k+1} + \sum_{i=1}^N h_i(t^{k+1}) \quad (6)$$

$$= \sigma^k + \tilde{E} \Delta \epsilon + \sum_{i=1}^N (\exp(-\Delta t/\tau_i) - 1) h_i(t^k) \quad (7)$$

where

$$\tilde{E} = E_\infty + \sum_{i=1}^N E_i \frac{\tau_i}{\Delta t} (1 - \exp(-\Delta t/\tau_i)).$$

Moving to 3D

$$G(t) = G_\infty + \sum_{i=1}^N G_i \exp(-t/\tau_i), \quad K(t) = K_\infty$$

$$\mathbf{h}_i(t^{k+1}) = \exp(-\Delta t/\tau_i) \mathbf{h}_i(t^k) + 2G_i \frac{\tau_i}{\Delta t} (1 - \exp(-\Delta t/\tau_i)) \Delta \mathbf{e}$$

where

$$\mathbf{e} = \boldsymbol{\epsilon} - \frac{1}{3} \nabla \cdot \mathbf{u} \mathbf{I}, \quad \Delta \mathbf{e} = \mathbf{e}^{k+1} - \mathbf{e}^k$$

As before we can define

$$\tilde{G} = G_\infty + \sum_{i=1}^N G_i \frac{\tau_i}{\Delta t} (1 - \exp(-\Delta t/\tau_i)).$$

and letting $\boldsymbol{\sigma} = \tilde{\mathbb{C}} \boldsymbol{\epsilon}$ we have

$$\tilde{\mathbb{C}} = K_\infty (\mathbf{I} \otimes \mathbf{I}) + 2\tilde{G} \left(\mathbb{I} - \frac{1}{3} \mathbf{I} \otimes \mathbf{I} \right)$$

The stress update finally becomes

$$p^{k+1} = K_\infty \text{Tr}(\boldsymbol{\epsilon}^{k+1})$$

and

$$\mathbf{s}^{k+1} = 2G_\infty \mathbf{e}^{k+1} + \sum_{i=1}^N \mathbf{h}_i^{k+1}$$

and

$$\boldsymbol{\sigma}^{k+1} = \mathbf{s}^{k+1} + p^{k+1} \mathbf{I}.$$

3 For Kaj - Where does the hereditary integral go?

Elastic

$$\tau_i(t) = \tau_{0,i} + \sum_{j=1}^N K_{ij}(t) [\delta_j(t) - V_{pl,j} t] - \eta V_i(t)$$

where K_{ij} is the dense GF matrix, i is the slip patch index.

For a visco-elastic case, the interaction kernel K_{ij} is replaced by a time-dependent kernel $K_{ij}(t - t')$, resulting in the hereditary integral

$$\tau_i(t) = \tau_{0,i} + \sum_{j=1}^N \int_0^t K_{ij}(t - t') \frac{\partial \delta_j(t')}{\partial t} dt' - \eta V_i(t) \quad (8)$$

where $\frac{\partial \delta_j(t')}{\partial t}(t') = V_j(t')$ is the slip velocity history of the patch. We note the $V_{pl,j}t$ will reappear in the background stress.

Correction (v5). The kernel $K_{ij}(t - t')$ in Eq. (8) is the *relaxation function*: the stress at patch i at time $t - t'$ after a unit of slip imposed at patch j and then held. Convolved with the slip rate as in Eq. (8) it integrates the slip history, which is the statement that the medium relaxes accumulated slip; it contains no Dirac part, the elastic response being its $t = 0$ value, $K_{ij}(0) = K_{ij}^{\text{elastic}}$. The correspondence principle of section 4 gives it exactly, with geometry-independent rates,

$$K_{ij}(t - t') = \sum_{p=1}^3 C_{ij}^{(p)} \exp\left(-\frac{t - t'}{\tau_p}\right) + C_{ij}^{(\infty)}, \quad \tau_m = \eta_{\text{visc}}/G, \quad (9)$$

with

$$\tau_1 = \tau_m, \quad \tau_2 = \frac{3(1 - \nu)}{1 + \nu} \tau_m, \quad \tau_3 = \frac{3}{2(1 + \nu)} \tau_m,$$

($1.8\tau_m$ and $1.2\tau_m$ at $\nu = 1/4$), $\sum_p C_{ij}^{(p)} + C_{ij}^{(\infty)} = K_{ij}^{\text{elastic}}$, and, for the homogeneous Maxwell half space, $C_{ij}^{(\infty)} = 0$ for the stress kernels (complete relaxation); full-space kernels need only τ_1, τ_2 . This was verified against the **interact** kernels at rounding level, with held-out samples that would expose a missing pole; no repeated-pole ($t e^{-t/\tau}$) terms occur. The amplitudes $C_{ij}^{(p)}$ need no new Green's functions: they follow from evaluating the elastic kernels at a few sampled effective moduli and solving a small weight system. The v4 form $K_{ij}^{\text{elastic}} \delta_{\text{dirac}} + K_{ij}^{\text{viscous}} e^{-(t-t')/\tau_m}$ is recovered in spirit only in the antiplane limit, where Eq. (9) reduces to the single term $K_{ij}^{\text{elastic}} e^{-(t-t')/\tau_m}$.

Evaluating the full hereditary integral at every time step requires saving the slip velocity history of all patches for all past time steps. This scales as $\mathcal{O}(N \times M)$ where M is the number of time steps. Fixing a temporal window of length $m' < M$ steps obviously reduces (and limits) the required memory.

Since the Maxwell relaxation follows an exponential decay $\exp(-t/\tau_M)$, you can avoid storing the entire history by introducing a viscous stress state variable, $H_{ij}(t)$, for every patch interaction

$$\tau_i(t) = \tau_{0,i} + \sum_{p=1}^3 h_i^{(p)}(t) + \sum_{j=1}^N C_{ij}^{(\infty)} [\delta_j(t) - V_{pl,j}t] - \eta V_i(t), \quad (10)$$

where the per-interaction H_{ij} of v4 aggregate to one state per patch and pole, $h_i^{(p)} = \sum_j H_{ij}^{(p)}$, because the decay rates are shared by all interactions ($C^{(\infty)}$ vanishes in the homogeneous case and is kept for the layered one). Instead of integrating over past history, you update H_{ij} dynamically at each time step using its differential form

$$\frac{dh_i^{(p)}}{dt} = -\frac{h_i^{(p)}}{\tau_p} + \sum_{j=1}^N C_{ij}^{(p)} [V_j(t) - V_{pl,j}], \quad (11)$$

and the stressing rate follows by differentiating Eq. (10),

$$\frac{d\tau_i}{dt} = \sum_{j=1}^N K_{ij}^{\text{elastic}} [V_j - V_{pl,j}] - \sum_{p=1}^3 \frac{h_i^{(p)}}{\tau_p} - \eta \dot{V}_i,$$

using $\sum_p C^{(p)} + C^{(\infty)} = K^{\text{elastic}}$: the cost over the elastic problem is three state vectors and three extra operator applications, with no history storage. If you didn't want to solve this additional ODE for the stress history, you could make an approximation by assuming that $V_j(t)$ is constant over a time-step Δt and then just directly integrate the LHS and RHS w.r.t. time t yielding

$$h_i^{(p)}(t + \Delta t) = h_i^{(p)}(t) \exp\left(-\frac{\Delta t}{\tau_p}\right) + \left[\sum_{j=1}^N C_{ij}^{(p)} \Delta \delta_j\right] \left(\frac{1 - \exp(-\Delta t/\tau_p)}{\Delta t/\tau_p}\right),$$

where $\Delta \delta_j = \hat{V}_j \Delta t$ and $\hat{V}_j$ is the assumed constant value of $V_j(t)$ over time $[t, t + \Delta t]$.

To see where this comes from, consider

$$\frac{dH}{dt} = -\frac{H}{\tau} + KV, \quad KV = \text{const.}$$

$$H(t) = KV\tau + (H_0 - KV\tau)e^{-t/\tau} = H_0e^{-t/\tau} + KV\tau(1 - e^{-t/\tau})$$

$$H(t) = H_0e^{-t/\tau} + K\frac{S}{\Delta t}\tau(1 - e^{-t/\tau})$$

To move from $t = 0$ where H_0 is defined, we evaluate the above equation at $t = \Delta t^{(1)} = t^{(1)}$ to give H_1 we have

$$H(\Delta t^{(1)}) = H_1 = H_0e^{-\Delta t^{(1)}/\tau} + K\frac{S}{\Delta t^{(1)}}\tau(1 - e^{-\Delta t^{(1)}/\tau})$$

The next step $t^{(2)} = t^{(1)} + \Delta t^{(2)}$ is given by

$$H(t^{(2)}) = H(t^{(1)} + \Delta t^{(2)}) = H_2 = H_1e^{-\Delta t^{(2)}/\tau} + K\frac{S}{\Delta t^{(2)}}\tau(1 - e^{-\Delta t^{(2)}/\tau}).$$

Note that each time-step k can be thought as a single integration moving from a reference state k at time $t^{(k)}$ to a new state which is advanced by $\Delta t^{(k)}$. For example at step 1 we move from $t = 0$ (reference state) to $t = \Delta t^{(1)}$ (new state) and for step 2 we move from state $t = t^{(1)}$ (reference state) to the state at $t = t^{(2)} = t^{(1)} + \Delta t^{(2)}$.

Derivation With the relaxation kernel of Eq. (9) inserted into Eq. (8) there is no Dirac term to integrate separately; each exponential contributes one history integral of the v4 form. Let (per pole p , writing $C = C_{ij}^{(p)}$ and $\tau = \tau_p$, and with V_j standing for $V_j - V_{pl,j}$)

$$\begin{aligned} H_{ij}^{(p)}(t) &= \int_0^t C \exp\left(-\frac{t-t'}{\tau}\right) V_j(t') dt' \\ &= \exp\left(-\frac{t}{\tau}\right) \int_0^t C \exp\left(\frac{t'}{\tau}\right) V_j(t') dt' \end{aligned} \quad (12)$$

Now compute the time derivative of $H_{ij}^{(p)}(t)$

$$\frac{dH}{dt} = \frac{d}{dt} \left[\exp\left(-\frac{t}{\tau}\right) \int_0^t C \exp\left(\frac{t'}{\tau}\right) V_j(t') dt' \right] \quad (13)$$

$$= \frac{d}{dt} [A(t)B(t)] \quad (14)$$

$$= -\frac{1}{\tau} \exp(-t/\tau) B(t) + A(t) \frac{d}{dt} \left[\int_0^t C \exp\left(\frac{t'}{\tau}\right) V_j(t') dt' \right] \quad (15)$$

$$= -\frac{1}{\tau} \exp(-t/\tau) B(t) + A(t) C \exp\left(\frac{t}{\tau}\right) V_j(t) \quad (16)$$

$$\frac{dH}{dt} = -\frac{1}{\tau} \left[\exp(-t/\tau) \int_0^t C \exp\left(\frac{t'}{\tau}\right) V_j(t') dt' \right] + CV_j(t) \quad (17)$$

$$= -\frac{1}{\tau} \left[H_{ij}^{(p)}(t) \right] + CV_j(t) \quad (18)$$

The essential idea is that if you have an ODE, say

$$\frac{dy}{dt} = F(t), \quad y(0) = y_0$$

you can obtain the value $y(T)$ via:

1. solving the ODE analytically and evaluating the solution $y(t)$ at $t = T$, or by;
2. computing the integral

$$y(T) = y_0 + \int_0^T F(t') dt'.$$

This is the reason why methods which are used to solve ODEs are often called “integrators”.

4 The Laplace Transform way

Basic idea

Use the correspondence principle: This is done in steps (1) take the v-constitutive equation (2) takes its Laplace transform (3) manipulate the expression so that it's form matches the form used in pure elasticity, this will entail defining Laplace domain dependent parameters e.g. $(G(s))$ (5) compute all Lamé parameters required by Green's function using $\bar{G}(s)$, K , (6) insert $\bar{\mu}(s)$, $\bar{\lambda}(s)$ in the Green's function, (7) transform back into the time domain.

Form 1 Start with the constitutive equation for an elastic material

$$\sigma_{ij} = 2G(\epsilon_{ij} - \frac{1}{3}\epsilon_{kk}\delta_{ij}) + K\epsilon_{kk}\delta_{ij} \quad (19)$$

$$= 2G\epsilon_{ij} + (K - \frac{2G}{3})\epsilon_{kk}\delta_{ij} \quad (20)$$

$$= 2G\epsilon_{ij} + \lambda\epsilon_{kk}\delta_{ij} \quad (21)$$

where $\lambda = K - 2G/3$.

Start with the constitutive equation for a visco-elastic material

$$\frac{d\sigma_{ij}}{dt} + \frac{1}{\tau_m}(\sigma_{ij} - \frac{1}{3}\sigma_{kk}\delta_{ij}) = 2G\frac{d\epsilon_{ij}}{dt} + \lambda\frac{d\epsilon_{kk}}{dt}\delta_{ij}.$$

Denote the Laplace transform of a function $f(t)$ as

$$\mathcal{L}\{f(t)\} = \bar{f}(s).$$

Basic function transformations:

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\bar{f}(s) - f(0), \quad \mathcal{L}\{c\} = \frac{c}{s}.$$

Take Laplace transform of the constitutive equation above to give

$$s\bar{\sigma}_{ij}(s) + \frac{1}{\tau_m}(\bar{\sigma}_{ij}(s) - \frac{1}{3}\bar{\sigma}_{kk}(s)\delta_{ij}) = 2Gs\bar{\epsilon}_{ij}(s) + \lambda s\bar{\epsilon}_{kk}(s)\delta_{ij}.$$

Do some algebra

$$\begin{aligned} (s + \frac{1}{\tau_m})\bar{\sigma}_{ij}(s) &= 2Gs\bar{\epsilon}_{ij}(s) + \lambda s\bar{\epsilon}_{kk}(s)\delta_{ij} + \frac{1}{\tau_m}\frac{1}{3}\bar{\sigma}_{kk}(s)\delta_{ij} \\ \bar{\sigma}_{ij}(s) &= 2G\frac{s}{s + \frac{1}{\tau_m}}\bar{\epsilon}_{ij}(s) + \frac{1}{s + \frac{1}{\tau_m}}(\lambda s)\bar{\epsilon}_{kk}(s)\delta_{ij} + \frac{1}{s + \frac{1}{\tau_m}}\left(\frac{1}{\tau_m}\frac{1}{3}\right)\bar{\sigma}_{kk}(s)\delta_{ij}, \end{aligned}$$

and then define

$$\bar{G}(s) = \frac{Gs}{s + \frac{1}{\tau_m}}.$$

We also have $K(s) = K$ because the material is not compressible in bulk.

Note to self. The Laplace transform of Form 1 introduces a term σ_{kk} which is not naturally present in the form given by Form 1. This term ONLY affects the volumetric stress. Hence Form 1 is going to be fine if you only intend to use the GF for the deviatoric stress (i.e. as required for the shear stress on the fault).

The Green's functions we work with are usually going to be written in terms of λ and ν , so we will need their form in the Laplace domain, e.g.

$$\begin{aligned} \bar{\lambda}(s) &= K - \frac{2}{3}\bar{G}(s), \\ \bar{\nu}(s) &= \frac{\bar{\lambda}(s)}{2(\bar{\lambda}(s) + \bar{\mu}(s))} = \frac{3K - 2\bar{\mu}(s)}{2(3K + \bar{\mu}(s))} \end{aligned}$$

Write GF in the following form decomposed form which isolates spatial terms and material properties associated with either the deviatoric stress and volumetric stress

$$\sigma(G, \nu, \delta(t)) = F_1(x)G\delta(t) + F_2(x)F_2^m(G, \nu)\delta(t)$$

in which $F_1(x)G$ describes the deviatoric part of the stress and $F_2(x)F_2^m(G, \nu)$ describes the volumetric part of the stress. Folks usually employ this definition

$$F_1(x)G = K^{\text{elastic}}(x),$$

that $\tau(t) = K^{\text{elastic}}(x)\delta(t)$. Transform to the Laplace domain

$$\bar{\sigma}(\bar{G}(s), \bar{\nu}(s), \bar{\delta}(s)) = F_1(x)\bar{G}(s)\bar{\delta}(s) + F_2(x)\bar{F}_2^m(\bar{G}(s), \bar{\nu}(s))\bar{\delta}(s)$$

Time domain

$$\tau(t) = K^{\text{elastic}}\Delta\delta(t)$$

Correction (v5). The correspondence substitution acts on the transform of the *slip*, $\bar{\delta}(s)$, not of the velocity (the two differ by a factor s , since $\bar{\delta} = \bar{V}/s$ for $\delta(0) = 0$); v4 carried $\bar{V}$ through this chain, which is where the velocity form of section 3 (v4) originated. Corrected:

$$\bar{\tau}(s) = \frac{\bar{G}(s)}{G}K^{\text{elastic}}\bar{\delta}(s)$$

Using $s/(s+a) = 1 - a/(s+a)$ we have

$$\bar{\tau}(s) = \left(1 - \frac{1/\tau_m}{s + 1/\tau_m}\right) K^{\text{elastic}}\bar{\delta}(s).$$

Collectively this leads to the following Laplace transform for the complete constitutive equation

$$\bar{\tau}(s) = K^{\text{elastic}}\bar{\delta}(s) - \left(\frac{1/\tau_m}{s + 1/\tau_m}\right) K^{\text{elastic}}\bar{\delta}(s).$$

Inverse Laplace Transform

$$\mathcal{L}^{-1}\{K^{\text{elastic}}\bar{\delta}(s)\} = K^{\text{elastic}}\delta(t)$$

Notes: Inverse transform of $1/(s+a)$ is e^{-at} , so inverse transform of $a/(s+a)$ is ae^{-at} . Multiplication of two functions of s is Laplace domain, i.e. $\bar{f}(s)\bar{g}(s)$ maps to convolution integral in the time domain. Hence we have

$$\mathcal{L}^{-1} \left\{ \left(\frac{1/\tau_m}{s + 1/\tau_m} \right) \bar{\delta}(s) \right\} = \int_0^t \frac{1}{\tau_m} \exp \left(\frac{-(t-t')}{\tau_m} \right) \delta(t') dt'.$$

Finally, for the complete constitutive equation we have

$$\tau(t) = K^{\text{elastic}} \delta(t) - \int_0^t \left(\frac{1}{\tau_m} K^{\text{elastic}} \right) \exp \left(\frac{-(t-t')}{\tau_m} \right) \delta(t') dt',$$

which, integrating by parts with $\delta(0) = 0$, is the step-response convolution of section 3,

$$\tau(t) = \int_0^t K^{\text{elastic}} \exp \left(\frac{-(t-t')}{\tau_m} \right) V(t') dt' :$$

the accumulated slip relaxes; the stress is not proportional to the instantaneous velocity. (This single- τ_m form is the antiplane limit of this decomposition, in which $F_2 \equiv 0$; the general kernel carries the additional poles below.) Wrapping this all up for an actual simulation, we can use Eq. (10) along with the ODEs given by Eq. (11), where the definition of the states is given by Eq. (12). For the general (plane strain and 3-D) kernels the same steps applied to the full decomposition $F_1(x)G + F_2(x)F_2^m(G, \nu)$ produce the three poles of Eq. (9): the F_1G part relaxes as $\bar{G}(s)/G$ and contributes only $\tau_1 = \tau_m$, while the s dependence of $F_2^m(\bar{G}(s), \bar{\nu}(s))$, whose G -carrying denominators are $3K + 4\bar{G}$ and $3K + \bar{G}$, contributes τ_2 and τ_3 .

Form 2 Elastic:

$$\tau_{ij} = 2G e_{ij}, \quad \frac{1}{3}\sigma_v = \frac{1}{3}K \epsilon_v \quad \longrightarrow \quad \sigma_{ij} = \tau_{ij} + \sigma_v \delta_{ij}$$

Visco-elastic with dev-vol parts

$$\frac{d\tau_{ij}}{dt} + \frac{1}{\tau_m} \tau_{ij} = 2G \frac{de_{ij}}{dt}$$

$$\frac{d\sigma_v}{dt} = K \frac{d\epsilon_v}{dt}$$

Laplace transform

$$\begin{aligned}
s\bar{\tau}_{ij}(s) + \frac{1}{\tau_m}\bar{\tau}_{ij} &= 2Gs\bar{e}_{ij}(s) \\
\bar{\tau}_{ij}(s) &= 2G\frac{s}{s + 1/\tau_m}\bar{e}_{ij}(s) \\
s\bar{\sigma}_v(s) &= Ks\bar{\epsilon}_v(s)
\end{aligned}$$

Results from the implementation (v5)

Three further results are recorded here for completeness; the numerical evidence is in the cited documents. First, displacement kernels carry no τ_m pole (the shear modulus cancels in displacements from dislocations, which depend on the moduli only through ν) and relax to the $\nu = 1/2$ field, the permanent deformation, so their constant term $C^{(\infty)}$ is nonzero; this is one reason the representation carries $C^{(\infty)}$ throughout. Second, a vertical (dip 90) fault appears to have ν -independent surface displacements and hence shows no postseismic surface deformation at all in a uniform Maxwell medium; dipping faults, finite-length strike-slip end effects, or layering are needed for a surface transient. Third, for an elastic plate of thickness H over a Maxwell half space of equal rigidity, the interface reflection coefficient in the Laplace domain is

$$\bar{\beta}(s) = \frac{G - \bar{G}(s)}{G + \bar{G}(s)} = \frac{a}{s + a}, \quad a = \frac{1}{2\tau_m},$$

so the image at depth shift $2nH$ carries $\bar{\beta}^n$, whose step response is $1 - e^{-at} \sum_{k < n} (at)^k / k!$: polynomial-times-exponential terms with clock $2\tau_m$. These are not representable exactly by finitely many simple exponentials, so in the layered case Eq. (9) becomes a controlled fit whose term count is measured against the image-series closed form; the state equations (10) and (11) are unchanged, with fitted rates and a nonzero relaxed operator (a plate over a substrate does not fully relax, which is what keeps cycle loading alive there).

Verification status (v6)

For the record, the state of testing as of this date; details and numbers live in `rsf_ve_test_ledger.md`. Exact and verified at rounding: the three-pole representation on all kernel families (`ve_check`), time histories against numerical transform inversion (`ve_laplace_check`), the prescribed-slip testbed

(`relax_fault_ve`), and the 2-D antiplane layered machinery against the Rybicki and Nur–Mavko closed forms, with a measured term-count curve (six terms reach 10^{-5} of coseismic). Verified empirically for the 3-D plate over Maxwell case (propagator reference): one shared six-rate ladder fits surface responses across mechanisms, depths, and stations at a median of 10^{-4} of signal amplitude. Negative results on record: the scalar image construction of section 4’s layered form does not transfer to 3-D (it recovers only a few percent to two thirds of the true transient), and rate ladders placed on the propagator’s root branches underperform blind geometric ones. In progress: interior stresses from the extended propagator, validated at the finite-difference floor for azimuthal orders 0 and 2 and currently failing for $|M| = 1$ content, which gates the 3-D layered stress kernels that the cycle solver needs. The uniform Maxwell case of sections 3 and 4 requires none of this: its interior stresses are exact by the equations above.